



SCHOOL OF PROGRAMMING AND DEVELOPMENT

Java Web Developer

Syllabus

UDACITY.COM

Overview

Advance your Java career with projects in web services, security, persistence, and DevOps with this Nanodegree. Learn industry-standard tools like Spring Boot, Jenkins, and Selenium used in modern enterprise development.

Nanodegree Program



Intermediate



69 hours



4.6 (294 Reviews)

Prerequisites

Prior to enrolling, you should have the following knowledge:

Basic SQL

Intermediate Java

You will also need to be able to communicate fluently and professionally in **written and spoken English**.

Skills You'll Learn

Spring Boot Basics [Java]

Selenium | Spring boot configuration | Object-relational mapping | Spring mvc | Servlets | Spring managed components | Spring security | Unit testing | Mybatis | Thymeleaf | Model-view-controller | Junit | Java testing | Functional testing | Application servers | Maven | Java package management | HTML template creation | Dependency injection | Java annotations | Java assertions | SQL queries | Behavioral software testing | Hashing | Spring boot | Data encryption | IntelliJ | Spel

Web Services and APIs

GraphQL | Swagger | REST APIs | Web services | Spring security | Microservices | Software testing | Application security basics | Spring boot | Soap | Microservices architecture basics | Front-end development basics | Data format fundamentals | JSON | API documentation | HTTP | XML | API security | API development | Integration testing | Unit testing | Errors and exceptions

Data Stores & Persistence

Object-relational mapping | Spring data jpa | Jpa | Unit testing | Multitier architecture | Hibernate ORM | Repository design pattern | Spring boot | Jpql | DAOs | SQL queries | Java data types | Relational database transactions | Spring boot configuration | Jdbc | Dtos | SQL scripts | Java inheritance | Spring managed components | Dependency injection

Security and DevOps

Docker | Git | Splunk | Logging | Application security best practices | CI/CD | Code coverage | Application security basics | Jenkins | Unit testing | Junit | Negative testing | Jwts | Happy path testing | Hashing | Application metrics



Courses

01. Welcome to Java Web Developer Nanodegree Program

- L1 Java Web Developer Nanodegree Orientation
- L2 Knowledge, Community, and Careers

02. Spring Boot Basics [Java]

25 hours

Learn the fundamentals of Java while being introduced to a Spring Boot framework and associated integrations and plugins.

- | | | |
|----|--|--|
| L1 | Introduction to Spring Boot | In this lesson, you will meet your instructor, learn about the prerequisites, and discuss what web development is and how it has been accomplished. |
| L2 | Web Development in Java | In this lesson, we're going to take a look at the structure of web applications in Java, Spring's role in web development, and the use of Maven as a dependency management tool. |
| L3 | Spring Boot Basics for Web Development | In this lesson, we'll be learning how Spring framework operates and how to integrate with it to build flexible, maintainable application components. |
| L4 | Spring MVC and Thymeleaf | In this lesson, we're going to learn how to build web sites with Spring! We'll be using Spring MVC and Thymeleaf. |
| L5 | Data Persistence & Security | In this lesson, we will explore connecting our Spring Boot applications to SQL databases and the benefits of persistent data. We'll be using MyBatis and Spring Security! |
| L6 | Testing | In this lesson, you'll learn how to use Selenium, write tests with JUnit, and integrate Selenium in order to test complex, multi-action user stories. |
| L7 | Project: SuperDuperDrive | |

03. Web Services and APIs

16 hours

Explore the differences between web services, APIs, and microservices. Develop REST and GraphQL APIs, and learn how to secure, consume, document, and test those APIs and web services.

L1	Overview	First, meet your instructor Kesha and learn about the course. Then, learn a brief overview of web services and how they compare to APIs and microservices.
L2	REST APIs	Learn how to build RESTful APIs with Spring Boot, first by learning about the REST architecture, then how to use annotations with REST controllers in Spring Boot.
L3	GraphQL APIs	Dive into GraphQL, a query language for APIs. GraphQL lets calling clients limit the data response to just what is needed.
L4	Microservices	Find out how Microservices improve on the older monolithic application structure, including how to use Eureka and Spring Data REST for easy development and deployment.
L5	Security	Security is key to keeping your information safe - here, you'll learn how to add security to your APIs to keep out unwanted eyes and avoid undesired changes.
L6	Consuming SOAP & REST	Enable the consumption of SOAP and REST APIs within your own API to further speed up development time and avoid reinventing the wheel.
L7	Documentation	Learn about the importance of documentation and easily implement it with the use of Swagger within your code.
L8	Unit & Integration Tests	Build unit and integration tests to give you confidence your application continues to work as intended after making new changes to your code.
L9	Project: Build the Backend System for a Car Website	In this project, you'll create a REST API to store vehicle data, while consuming information regarding location and pricing data from external resources. Also, you'll document and test your API!

04. Data Stores & Persistence

18 hours

Build applications that read and write to relational databases using both the Java Persistence API (JPA) and SQL. Use standard design patterns to make your persistence layer easy to test and integrate with a Spring Boot application.



L1	Introduction to Data Stores & Persistence	An introduction to the Data Stores & Persistence course. You will learn about persisting data in web-based Java applications.
L2	Data in Multitier Architecture	This lesson discusses Entity design and the relationship between data in the various layers of your application.
L3	Java Persistence API (JPA)	We'll be looking at using JPA to manage the state of our Entities for us. We'll learn about the concept of a Persistence Context and using an Entity Manager to change the state of our Entities.
L4	Connecting To Data Sources	In this lesson, you'll set up an external database, connect to external databases, configure and initialize multiple data sources, and configure data sources for unit tests.
L5	Persistence Without JPA	In this lesson, we'll be covering initialization with SQL, Data Objects vs. Entities, Data Access Objects (DAOs), JdbcTemplate, and Performance.
L6	Project: Critic Chronologer	

05. Security and DevOps

11 hours

Learn about Git, version control and best practices for authorization and authentication. Use Jenkins to build a CI/CD pipeline to deploy code to production.

L1	Authentication and Authorization	Implement authentication on a Spring Boot application using a JSON Web Token (JWT).
L2	Testing and Logging	Refresh testing fundamentals. Use logging to get insights into your code and debug your applications. Get practice with two commonly used logging utilities: SLF4J and Log4J.
L3	Splunk	Splunk was founded in 2003 to make sense of machine-generated data. Here you'll learn about what Splunk can be used for, how to install it, and how to use it for generating dashboards and more.
L4	CI&CD	
L5	Project: eCommerce Application	



06. Intro to Java Programming

9 hours

L1	Variables and Data Types	Discover the basic idea behind programming and write your first Java program. Use variables to store and retrieve information.
L2	Control Flow and Conditionals	Use if-else statements and switch-case statements to write Java programs to tackle any decision making scenario.
L3	Functions	Create functions in Java. Declare methods with input arguments and return types. Then, call the functions you create.
L4	Loops	Use for and while loops to iterate over a block of code. Learn how to write nested loops. Use 2D arrays to store and retrieve information.
L5	IntelliJ and Debugging	Prepare for your next project by installing Java and IntelliJ, debugging your code, and applying the skills learned in previous lessons.

Meet Your Instructors



Peter Zastoupil

Instructor

Peter Zastoupil is an enterprise developer and technical administrator. He has seven years of on-the-job experience building features for massive enterprise Java servers, and over four years of teaching those skills to new developers.



Alex Pritchard

Senior Software Engineer

Alex is a Senior Software Engineer for CPA Global. He is excited to combine his background as a music educator with more than a decade of enterprise Java experience to help create this hands-on course about testing and deploying Java applications.



Kesha Williams

Instructor

Kesha has over 20 years experience in software development and is a software engineering manager at Chick-fil-A, routinely leading innovation teams in proving out the use of cloud services to solve complex business problems. She was recently named an Alexa Champion by Amazon.



Sareeta Panda

Instructor

Sareeta is a Java enthusiast and Senior Developer at Walmart e-Commerce. She specializes in Enterprise Application development with Java and Kafka, NoSQL, Spring security, and CI/CD. Sareeta has over a decade of experience, spanning recently acquired startups to top Fortune 500 companies.

Why Udacity



Demonstrate proficiency with practical projects

Projects are based on real-world scenarios and challenges, allowing you to apply the skills you learn to practical situations, while giving you real hands-on experience

- ✓ Gain proven experience
- ✓ Retain knowledge longer
- ✓ Apply new skills immediately



24/7 access to real human support

Reviewers provide timely and constructive feedback on your project submissions, highlighting areas of improvement and offering practical tips to enhance your work

- ✓ Get help from subject matter experts
- ✓ Gain valuable insights and improve your skills
- ✓ Learn industry best practices